

SymBioForge

Autonomous Circular Manufacturing Intelligence

NitroStack MCP Server

Built for the NitroStack Hackathon
Manufacturing & Industry 4.0 Track

Team: 4 members from Amrita University

July 2026

1. What is SymBioForge?

SymBioForge is an AI-powered platform that automatically connects factories so that one factory's waste becomes another factory's raw material. Think of it as a **matchmaker for industrial waste**.

Imagine a textile mill that produces cotton lint as waste. A nearby paper mill could use that cotton lint as raw material. SymBioForge finds these connections automatically, calculates how much CO₂ is saved, how much money each factory earns, and even generates government compliance reports.

The Simple Idea

Instead of factories dumping waste in landfills (which pollutes land, water, and air), SymBioForge creates a **circular economy** where waste flows from one factory to another, reducing pollution and creating economic value. This is called **Industrial Symbiosis**.

What Makes It Special

- **Fully Autonomous** -- 8 AI agents work together without any human clicking buttons
- **Self-Healing** -- if a factory shuts down, the system automatically finds alternative waste routes
- **Real Data** -- uses 15 real factories from Coimbatore, India with actual waste types
- **Government Ready** -- generates real SPCB (State Pollution Control Board) compliance PDF reports
- **Built on NitroStack** -- uses the Model Context Protocol (MCP) framework with visual widgets

2. How It Works (The Big Picture)

When the server starts, everything happens automatically in this order:

Step	What Happens	Who Does It
1	Load 15 factories from Coimbatore	System
2	Classify all waste streams (type, toxicity, reuse potential)	Profiler Agent
3	Infer hidden wastes based on industry type	Profiler Agent
4	Find all possible factory-to-factory waste matches	Matchmaker Agent
5	Discover multi-hop chains (A gives to B gives to C)	Matchmaker Agent
6	Invent new products from waste streams	Inventor Agent
7	Promote the best opportunities	Auditor Agent
8	Create step-by-step manufacturing blueprints	Architect Agent
9	Start monitoring for disruptions and compliance	Sentinel Agent
10	Drip-feed 3 more factories (1 per minute)	Scheduler

All of this happens in **under 2 seconds** at startup. No human touches anything. The judges just open the app and everything is already running.

3. The 8 Autonomous Agents

SymBioForge has 8 AI agents that communicate through an Event Bus (a message system). When one agent finishes its job, it publishes an event, and the next agent picks it up automatically.

1. Clerk Agent

The receptionist. Registers new factories into the system and generates SPCB Form V compliance PDFs. When a factory joins, the Clerk logs it and triggers the rest of the chain.

2. Scout Agent

The inspector. Profiles new or updated factories -- checks their raw materials, production capacity, and location. Prepares the factory data for the Profiler.

3. Profiler Agent

The waste scientist. Classifies every waste stream by type (organic, chemical, metallic, etc.), toxicity level, and reuse potential. Also INFERS hidden wastes -- e.g., a textile mill probably produces dye effluent even if they didn't declare it.

4. Matchmaker Agent

The matchmaker. The core engine. Uses Haversine distance (how far apart factories are) + compatibility scoring (can this waste actually be used?) + volume matching to find symbiotic pairs. Also discovers multi-hop chains: Factory A's waste goes to B, whose waste goes to C.

5. Inventor Agent

The product designer. Looks at available waste streams and invents new products. For example: rice husk + fly ash = eco-bricks. Uses manufacturing process databases and market data to estimate revenue.

6. Auditor Agent

The quality gate. Reviews all matches and products, scores them, and promotes the top ones to 'Blueprint Ready' status. Only the best opportunities move forward.

7. Architect Agent

The engineer. Takes promoted opportunities and creates detailed manufacturing blueprints: equipment needed, facility requirements, capital cost, payback period, and regulatory notes.

8. Sentinel Agent

The guardian. Runs every 30 seconds checking ecosystem health. Monitors waste volume changes (alerts if >25% deviation), tracks SPCB compliance deadlines (warns at 330 days, marks overdue at 365), and self-heals when a factory goes down.

4. The Event Chain

The agents communicate through events. Here is the main chain:

Event	Meaning	Triggers
FACTORY_REGISTERED	A new factory joined the cluster	Scout starts profiling
FACTORY_PROFILED	Waste streams have been classified	Matchmaker finds matches
MATCHES_DISCOVERED	Symbiotic pairs found	Inventor generates products
PRODUCTS_INVENTED	New products created from waste	Auditor evaluates all
IMPACT_AUDITED	Top opportunities promoted	Architect plans blueprints
PATHWAYS_DEIGNED	Blueprints are ready	Sentinel starts monitoring
ECOSYSTEM_STABLE	Everything is healthy	Sentinel keeps watching
SENTINEL_TRIGGERED	A disruption detected	Matchmaker re-routes waste
VOLUME_UPDATE	Waste volume changed	Sentinel checks baseline
COMPLIANCE_DUE	Filing deadline approaching	Clerk generates report

5. MCP Tools (What Users Can Do)

SymBioForge exposes 15 tools through the Model Context Protocol. Users interact by typing natural language in NitroStudio's AI Chat -- the AI picks the right tool automatically.

Tool	What It Does	Widget
get-cluster-state	See live state: factories, matches, products, logs	Agent Swarm Monitor
get-ecosystem-map	Visualize factory network with symbiotic edges	Ecosystem Map
get-carbon-metrics	CO2, water, energy saved + financial value	Carbon Dashboard
get-opportunity-feed	Ranked list of matches and products	Opportunity Feed
get-product-concepts	AI-invented products from waste	Product Cards
get-waste-profiles	Classified waste streams per factory	Waste Profiles
get-pathway	Step-by-step blueprint for an opportunity	Pathway Viewer
register-factory	Add a new factory (triggers full agent chain)	Compliance Dashboard
get-compliance-report	Generate SPCB Form V PDF	Compliance Dashboard
control-swarm	Start, stop, or reset the agent swarm	Swarm Monitor
trigger-disruption	Simulate factory shutdown (test self-healing)	Swarm Monitor
run-simulation	SymbioSim: 12-month time machine	Timeline
get-impact-story	Human-readable impact equivalencies	Impact Story
get-district-overview	SPCB officer governance dashboard	District Dashboard

6. The 4 Hackathon Stretch Goals

Beyond the core system, we built 4 additional features to wow judges:

Idea A: "Ask SymBioForge" AI Persona

Judges can have a conversation with the ecosystem! They ask questions like "Why did you pair the foundry with the glass works?" and the AI reads live cluster data -- factories, matches, chains, metrics -- and explains the chemistry, logistics, and financial reasoning behind every autonomous decision. No other team has an AI that explains its own decisions.

Idea B: SPCB District Dashboard

A governance view for District Environmental Officers. Shows cluster-wide compliance rates, factories approaching filing deadlines (red/yellow/green risk heatmap), landfill diversion progress against district targets, carbon credits earned, and industry sector breakdown. Positions SymBioForge as a government platform, not just a factory tool -- aligned with India's Smart Cities Mission.

Idea C: Impact Story

Translates raw metrics into human impact. Instead of '184 tons CO₂' (which means nothing to most people), it says '40 cars off the road for a year'. Instead of 'INR 22L saved', it says '11 MSME micro-loans funded'. Shows jobs created, tax revenue, land saved from contamination, and UN SDG alignment (Goals 9, 11, 12, 13). Turns technical data into a pitch deck that sells itself.

Idea D: SymbioSim Time Machine

The showstopper. Judges can watch 12 months of ecosystem evolution play out. The simulation resets the cluster, then month by month: factories join, matches form, products get invented, a factory has an emergency shutdown (disruption!), Sentinel self-heals, and the circular economy score climbs from 0% to ~78%. Every other team shows a static screenshot. SymBioForge shows a MOVIE.

7. Tech Stack

Layer	Technology	Purpose
Framework	NitroStack MCP (@nitrostack/core)	Model Context Protocol server
Language	TypeScript (strict, ES Modules)	Type-safe codebase
Validation	Zod schemas	Input validation on all tools
Widgets	Next.js 14 + @nitrostack/widgets	12 visual dashboard UIs
Icons	Lucide React	Consistent iconography
PDF	pdfkit	SPCB Form V compliance reports
Database	Supabase (optional)	Persistent factory storage
Transport	STDIO (dev) / HTTP (cloud)	MCP communication
Protocol	Model Context Protocol	AI-tool interoperability

8. Project Structure

Folder	Files	What It Contains
src/agents/	8 files	8 autonomous agents (clerk, scout, profiler, matchmaker, inventor, auditor, architect, sentinel)
src/core/	8 files	Business logic: waste classifier, compatibility matrix, matching algorithm, product generator, im
src/orchestrator/	4 files	Event bus (pub/sub), state manager (singleton), scheduler (drip feed + health checks), agent ch
src/modules/	15 files	15 MCP tool controllers + bootstrap + module registration + prompt resource
src/widgets/	12 folders	12 Next.js widget UIs for dashboards, maps, feeds, timelines
src/data/	7 files	JSON fixtures: factories, materials-db, compatibility matrix, emission factors, market data, manu

9. By the Numbers

Metric	Value
Total factories in cluster	18 (15 initial + 3 drip-fed)
Symbiotic matches discovered	57
AI-invented products	3
Manufacturing blueprints	5
Circular economy score	11%
CO2 avoided	184 tons/year
Water saved	1.46 million liters/year
Energy saved	291 MWh/year
Financial value generated	INR 22 lakhs/year
Landfill waste diverted	145.5 tons/year
Autonomous agents	8
MCP tools	15
Widget dashboards	12
Event types	11
Lines of code (approx)	5,000+

10. 5-Minute Demo Script

Step 1 (0:00): Open NitroStudio. Show the App Canvas -- 28 items: 15 tools, 13 resources, 1 prompt. Server is connected, STDIO transport. The ecosystem has already bootstrapped autonomously.

Step 2 (0:30): Open AI Chat. Type 'show me the cluster state'. The Agent Swarm Monitor widget appears showing 8 agents, 18 factories, 57 symbioses, live activity logs with Sentinel health checks running every 30 seconds.

Step 3 (1:00): Type 'show me the carbon metrics'. The Carbon Dashboard shows circular economy score, CO₂ avoided, water saved, energy saved, and financial value.

Step 4 (1:30): Type 'show me the impact story'. The Impact Story widget translates raw numbers into human impact: cars off the road, MSME loans funded, land saved, jobs created, SDG alignment.

Step 5 (2:00): Register a new factory. Type the factory details. Watch the Agent Swarm Monitor -- Clerk registers, Scout profiles, Profiler classifies, Matchmaker finds matches. Full chain fires in real time.

Step 6 (2:45): Type 'trigger disruption at factory fact_1'. Sentinel detects the shutdown, logs affected chains, re-triggers Matchmaker. Self-healing in action.

Step 7 (3:15): Type 'show me the district overview'. The SPCB Officer Dashboard appears: compliance rates, risk heatmap, deadline alerts, carbon credits, industry breakdown.

Step 8 (3:45): Type 'generate compliance report for fact_1'. A real SPCB Form V PDF is generated with factory profile, classified waste streams, and impact metrics.

Step 9 (4:15): Type 'run the simulation'. SymbioSim Time Machine plays 12 months: factories join, matches form, disruption hits, self-healing kicks in, score climbs to ~78%.

Step 10 (4:45): Q&A.; Judges can ask the AI anything: 'Why did you pair these factories?' 'Which industry should I add next?' The AI explains with real data.

11. Why SymBioForge Wins

1. Zero Human-in-the-Loop

Most teams build tools that require a human to click buttons and make decisions. SymBioForge's 8 agents work together autonomously -- the ecosystem self-organizes, self-heals, and self-improves.

2. Real-World Impact

This isn't a toy. It uses real factories from Coimbatore, real waste types from India's industrial registry, real SPCB Form V compliance formats, and real emission factors. It could be deployed tomorrow.

3. Multiple Stakeholders

We didn't just build for factories. We built for SPCB officers (District Dashboard), for investors (Impact Story), for city planners (Carbon Dashboard), and for the general public (SDG alignment). This is a platform, not a demo.

4. The AI Explains Itself

No other team has an AI that can explain its own autonomous decisions. Judges can interrogate SymBioForge during Q&A; and get data-driven answers about why specific factories were paired.

5. The Time Machine

Every other team shows a static snapshot. We show a MOVIE -- 12 months of an ecosystem coming alive, complete with a disruption crisis and self-healing recovery. It's unforgettable.

SymBioForge -- Turning Industrial Waste into Shared Prosperity

Built with NitroStack MCP by 4 students from Amrita University